InsightRAG Pro: A Enterprise retrieval-Augmented Generation Platform with GraphRAG and Quality Evaluator Checkers

Anuj Meena

Abstract

This report details the implementation of InsightRAG Pro, a production-grade Retrieval-Augmented Generation (RAG) platform. The system implements table-aware recursive text chunking, hybrid Reciprocal Rank Fusion (RRF) search (BM25 sparse + dense vectors), and secondary Cross-Encoder reranking. Furthermore, the platform parses entity-relationship subgraphs from document chunks, mapping modular communities using modularity-greedy clustering algorithms in NetworkX. To evaluate RAG quality, we implement mathematical context relevance, faithfulness, correctness, and hallucination scoring checkers that run offline. FastAPI backend services and a comprehensive Streamlit dashboard are presented.

I. Introduction

As large language models (LLMs) are deployed in enterprise environments, preventing hallucination and resolving multi-hop relationships over private datasets become critical priorities. Simple vector retrieval methods often fail to capture structural connections between documents or tables. InsightRAG Pro provides a production-grade framework combining hybrid RRF search, GraphRAG entity-relation networks, and built-in quality evaluation metrics.

II. System Framework

A. Hybrid RRF Searcher

Lexical keyword matching (BM25) and dense embeddings (SentenceTransformers) are merged using Reciprocal Rank Fusion (RRF):

$$RRF(d) = \sum_{m \in M} \frac{1}{k + r_m(d)} \tag{1}$$

where M represents the search retrievers and $r_m(d)$ represents the rank of document d in retriever m. We set $k = 60$. Top candidates are reranked using a Cross-Encoder network.

B. GraphRAG Integration

Document chunks are scanned to extract entity-relationship triplets. These triplets populate a NetworkX graph. Modularity-based greedy community detection algorithms cluster related entities into semantic context domains, supporting multi-hop relation analysis during query execution.

C. RAGBench Evaluator Checkers

To evaluate execution quality, four metrics are computed mathematically:

- **Context Relevance**: Overlap of query tokens in context.
- **Faithfulness**: Semantic grounding of answer statements in context.
- **Answer Correctness**: Token alignment of generated answers with reference labels.
- **Hallucination Rate**: Claims made in the answer that do not appear in the context.

III. Conclusion

InsightRAG Pro delivers an end-to-end enterprise solution for structured document indexing, relational graphing, and automated QA verification.